

# STAT 140: Design of Experiments

## Section 8 Solutions: Two Factors and Full Factorial Designs

### Exercise 1

Consider the observed outcomes from the completely randomized  $2^2$  factorial design with two replications per treatment combination.

| Unit ( $i$ ) | $(-1, -1)$ | $(-1, +1)$ | $(+1, -1)$ | $(+1, +1)$ | Assigned   |
|--------------|------------|------------|------------|------------|------------|
| 1            | ?          | 11         | ?          | ?          | $(-1, +1)$ |
| 2            | ?          | ?          | ?          | 13         | $(+1, +1)$ |
| 3            | 8          | ?          | ?          | ?          | $(-1, -1)$ |
| 4            | ?          | ?          | ?          | 11         | $(+1, +1)$ |
| 5            | ?          | 9          | ?          | ?          | $(-1, +1)$ |
| 6            | ?          | ?          | 9          | ?          | $(+1, -1)$ |
| 7            | 12         | ?          | ?          | ?          | $(-1, -1)$ |
| 8            | ?          | ?          | 7          | ?          | $(+1, -1)$ |

First compute the observed mean in each treatment combination:

$$\begin{aligned}\bar{Y}_{(-1,-1)}^{obs} &= \frac{8+12}{2} = 10, & \bar{Y}_{(-1,+1)}^{obs} &= \frac{11+9}{2} = 10, \\ \bar{Y}_{(+1,-1)}^{obs} &= \frac{9+7}{2} = 8, & \bar{Y}_{(+1,+1)}^{obs} &= \frac{13+11}{2} = 12.\end{aligned}$$

So the vector of treatment means is

$$\bar{Y}^{obs} = (10, 10, 8, 12).$$

### (a) Unbiased estimates of the three population-level factorial effects

**Main effect of  $F_1$ :**

$$\hat{\tau}_{F_1} = \frac{1}{2} \left[ \bar{Y}_{(+1,-1)}^{obs} - \bar{Y}_{(-1,-1)}^{obs} + \bar{Y}_{(+1,+1)}^{obs} - \bar{Y}_{(-1,+1)}^{obs} \right].$$

Substitute the values:

$$\hat{\tau}_{F_1} = \frac{1}{2} [8 - 10 + 12 - 10] = \frac{1}{2}(0) = 0.$$

**Main effect of  $F_2$ :**

$$\hat{\tau}_{F_2} = \frac{1}{2} \left[ \bar{Y}_{(-1,+1)}^{obs} - \bar{Y}_{(-1,-1)}^{obs} + \bar{Y}_{(+1,+1)}^{obs} - \bar{Y}_{(+1,-1)}^{obs} \right].$$

Substitute the values:

$$\hat{\tau}_{F_2} = \frac{1}{2} [10 - 10 + 12 - 8] = \frac{1}{2}(4) = 2.$$

**Interaction effect  $F_1 * F_2$ :**

$$\hat{\tau}_{F_1 * F_2} = \frac{1}{2} \left[ (\bar{Y}_{(+1,+1)}^{obs} - \bar{Y}_{(-1,+1)}^{obs}) - (\bar{Y}_{(+1,-1)}^{obs} - \bar{Y}_{(-1,-1)}^{obs}) \right].$$

Substitute the values:

$$\hat{\tau}_{F_1 * F_2} = \frac{1}{2} [(12 - 10) - (8 - 10)] = \frac{1}{2}[2 - (-2)] = \frac{1}{2}(4) = 2.$$

Therefore, the unbiased estimates are

$$\hat{\tau}_{F_1} = 0, \quad \hat{\tau}_{F_2} = 2, \quad \hat{\tau}_{F_1 * F_2} = 2.$$

## (b) Estimated main effects plot

To draw the main effects plot, compute the average response at each level of each factor.

**For factor  $F_1$ :**

$$\text{Average at } F_1 = -1 = \frac{\bar{Y}_{(-1,-1)}^{obs} + \bar{Y}_{(-1,+1)}^{obs}}{2} = \frac{10 + 10}{2} = 10,$$

$$\text{Average at } F_1 = +1 = \frac{\bar{Y}_{(+1,-1)}^{obs} + \bar{Y}_{(+1,+1)}^{obs}}{2} = \frac{8 + 12}{2} = 10.$$

So the main effects plot for  $F_1$  goes from  $(-1, 10)$  to  $(+1, 10)$ , which is a flat line.

**For factor  $F_2$ :**

$$\text{Average at } F_2 = -1 = \frac{\bar{Y}_{(-1,-1)}^{obs} + \bar{Y}_{(+1,-1)}^{obs}}{2} = \frac{10 + 8}{2} = 9,$$

$$\text{Average at } F_2 = +1 = \frac{\bar{Y}_{(-1,+1)}^{obs} + \bar{Y}_{(+1,+1)}^{obs}}{2} = \frac{10 + 12}{2} = 11.$$

So the main effects plot for  $F_2$  goes from  $(-1, 9)$  to  $(+1, 11)$ .

Interpretation:

- $F_1$  shows no average main effect because the mean response is 10 at both levels.
- $F_2$  has a positive main effect because the mean response increases from 9 to 11 when  $F_2$  changes from  $-1$  to  $+1$ .

**(c) Interaction plot with  $F_2$  on the horizontal axis**

Now hold  $F_1$  fixed and trace the response as  $F_2$  changes.

**Line for  $F_1 = -1$ :**

$$(F_2 = -1, \bar{Y}_{(-1,-1)}^{obs}) = (-1, 10), \quad (F_2 = +1, \bar{Y}_{(-1,+1)}^{obs}) = (+1, 10).$$

This is a horizontal line at height 10.

**Line for  $F_1 = +1$ :**

$$(F_2 = -1, \bar{Y}_{(+1,-1)}^{obs}) = (-1, 8), \quad (F_2 = +1, \bar{Y}_{(+1,+1)}^{obs}) = (+1, 12).$$

This line increases from 8 to 12.

Because the two lines are not parallel, there is interaction between  $F_1$  and  $F_2$ .

**(d) Interaction plot with  $F_1$  on the horizontal axis**

Now hold  $F_2$  fixed and trace the response as  $F_1$  changes.

**Line for  $F_2 = -1$ :**

$$(F_1 = -1, \bar{Y}_{(-1,-1)}^{obs}) = (-1, 10), \quad (F_1 = +1, \bar{Y}_{(+1,-1)}^{obs}) = (+1, 8).$$

This line decreases from 10 to 8.

**Line for  $F_2 = +1$ :**

$$(F_1 = -1, \bar{Y}_{(-1,+1)}^{obs}) = (-1, 10), \quad (F_1 = +1, \bar{Y}_{(+1,+1)}^{obs}) = (+1, 12).$$

This line increases from 10 to 12.

Again, the two lines are not parallel, confirming a positive interaction.

**(e) Optimal combination of factors**

To maximize the average response, compare the four treatment means:

$$\bar{Y}_{(-1,-1)}^{obs} = 10, \quad \bar{Y}_{(-1,+1)}^{obs} = 10, \quad \bar{Y}_{(+1,-1)}^{obs} = 8, \quad \bar{Y}_{(+1,+1)}^{obs} = 12.$$

The largest observed mean is 12, which occurs at treatment combination  $(+1, +1)$ .

The optimal factor combination is

$$F_1 = +1, \quad F_2 = +1.$$

## Exercise 2

### (a) Number of possible randomizations

There are 8 units total, with exactly 2 assigned to each of the four treatment combinations. The number of possible complete randomizations is:

$$\binom{8}{2} \binom{6}{2} \binom{4}{2} \binom{2}{2} = \frac{8!}{2!2!2!2!}.$$

Compute it:

$$\frac{8!}{2!2!2!2!} = \frac{40320}{16} = 2520.$$

There are 2520 possible randomizations.

### (b) How to generate the null randomization distribution of $\hat{\tau}_{F_1}$

Under the sharp null hypothesis, each unit has the same outcome under every treatment combination. That means the observed outcomes can be treated as fixed, and only the treatment assignment changes from one randomization to another.

To generate the null randomization distribution of  $\hat{\tau}_{F_1}$ :

1. Keep the eight observed outcomes fixed.
2. Reassign the eight units to the four treatment combinations, always keeping exactly two units in each combination.
3. For each reassignment, compute the treatment means

$$\bar{Y}_{(-1,-1)}^{obs}, \quad \bar{Y}_{(-1,+1)}^{obs}, \quad \bar{Y}_{(+1,-1)}^{obs}, \quad \bar{Y}_{(+1,+1)}^{obs}.$$

4. Use those means to calculate

$$\hat{\tau}_{F_1} = \frac{1}{2} \left[ \bar{Y}_{(+1,-1)}^{obs} - \bar{Y}_{(-1,-1)}^{obs} + \bar{Y}_{(+1,+1)}^{obs} - \bar{Y}_{(-1,+1)}^{obs} \right].$$

5. Repeat this over all possible randomizations, or over many simulated randomizations, and record the resulting values of  $\hat{\tau}_{F_1}$ .

The collection of these values forms the null randomization distribution of  $\hat{\tau}_{F_1}$ .

**(c) R code and observation****Exercise 3: Full Factorial Design**

Imagine a  $2^3$  experiment with factors  $F_1$ ,  $F_2$ , and  $F_3$ .

Each unit  $i$  has eight potential outcomes:

$$\begin{aligned} &Y_i(-1, -1, -1), Y_i(-1, -1, +1), Y_i(-1, +1, -1), Y_i(-1, +1, +1), \\ &Y_i(+1, -1, -1), Y_i(+1, -1, +1), Y_i(+1, +1, -1), Y_i(+1, +1, +1). \end{aligned}$$

To define the unit-level effect of  $F_1$ , first hold  $F_2$  and  $F_3$  fixed and compare the potential outcome at  $F_1 = +1$  to the potential outcome at  $F_1 = -1$ :

$$\begin{aligned} \tau_{i,F_1|F_2=-1,F_3=-1} &= Y_i(+1, -1, -1) - Y_i(-1, -1, -1), \\ \tau_{i,F_1|F_2=-1,F_3=+1} &= Y_i(+1, -1, +1) - Y_i(-1, -1, +1), \\ \tau_{i,F_1|F_2=+1,F_3=-1} &= Y_i(+1, +1, -1) - Y_i(-1, +1, -1), \\ \tau_{i,F_1|F_2=+1,F_3=+1} &= Y_i(+1, +1, +1) - Y_i(-1, +1, +1). \end{aligned}$$

Then average over the four possible  $(F_2, F_3)$  settings:

$$\begin{aligned} \tau_{i,F_1} &= \frac{1}{4} \left[ \tau_{i,F_1|F_2=-1,F_3=-1} + \tau_{i,F_1|F_2=-1,F_3=+1} + \tau_{i,F_1|F_2=+1,F_3=-1} + \tau_{i,F_1|F_2=+1,F_3=+1} \right] \\ &= \frac{1}{4} \left[ Y_i(+1, -1, -1) - Y_i(-1, -1, -1) + Y_i(+1, -1, +1) - Y_i(-1, -1, +1) \right. \\ &\quad \left. + Y_i(+1, +1, -1) - Y_i(-1, +1, -1) + Y_i(+1, +1, +1) - Y_i(-1, +1, +1) \right]. \end{aligned}$$

The population-level effect of  $F_1$  is the average of the unit-level effects across all units:

$$\tau_{F_1} = \frac{1}{N} \sum_{i=1}^N \tau_{i,F_1}.$$